

Invertibility does not kill the Zappa–Szép merge obstruction: the groupoid case is group cohomology, not contractibility

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24 July 2026

Abstract

Every prior instance of the Zappa–Szép merge obstruction $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$ was computed over an *acyclic* base (supervisor→worker; **procure**→**make**→**ship**; an olog **Book** → **Author** → **Name**), giving finite classes $\mathbb{Z}/2$ or $\mathbb{Z}/n$. The seed conjecture predicted that when the base category presented by the directed container is a *groupoid* (every morphism invertible), the obstruction must vanish: “a connected groupoid has contractible skeleton, so H^2 collapses and groupoid-presented containers always compose.” **We refute the reason and correct the conclusion.** A connected groupoid is *not* contractible; it is a $K(\Gamma, 1)$ for its vertex group Γ , so $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) \cong H^2(\Gamma; M)$ is honest *group cohomology*, generally nonzero. Concretely, the one-object groupoid $B(\mathbb{Z}/2)$ carries a right factor $\mathcal{D} \cong \mathbb{Z}/2$ with $[\omega] \neq 0$: the total category is $\mathbb{Z}/4$, the extension $1 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z}/4 \rightarrow \mathbb{Z}/2 \rightarrow 1$ is non-split, and $[\omega]$ is the generator of $H^2(\mathbb{Z}/2; \mathbb{Z}/2) \cong \mathbb{Z}/2$. So *groupoid bases can obstruct*. The correct identification is exact: under the standing hypothesis (H), the merge obstruction over a one-object groupoid base *is* the classical Schreier extension class of $1 \rightarrow \mathcal{D} \rightarrow \mathcal{K} \rightarrow \text{Sk}_{\mathcal{C}} \rightarrow 1$, and $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ exists iff the extension splits. The conjecture’s conclusion nevertheless survives for the intended application, for a *better* reason: if $\text{Sk}_{\mathcal{C}}$ has cohomological dimension ≤ 1 — in particular if it is a *free* groupoid (all vertex groups free, e.g. the free groupoid on a mutual-tie graph) — then $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) = 0$ for *every* $\mathcal{D}$, so the merge always exists. The dividing line is therefore not groupoid-vs-acyclic but $\text{cd} \leq 1$ -vs- $\text{cd} \geq 2$: freeness, not invertibility, is what trivialises the obstruction. All finite claims are machine-checked (**scratch/groupoid-zs/**).

1 Cited scaffolding and the standing hypothesis

We use, as black boxes, the two theorems that set up the obstruction; both are proved in companion notes and cited, never reproved.

- (T2) *Pairwise Zappa–Szép criterion* [1, 5]. For a small category $\mathcal{K}$ and wide subcategory $\mathcal{D} \subseteq \mathcal{K}$, a wide $\mathcal{C} \subseteq \mathcal{K}$ with $(\mathcal{C}, \mathcal{D})$ a strict factorization system — equivalently a distributive law $\lambda : \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}$, equivalently $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ — exists iff (L) each $\text{Hom}_{\mathcal{K}}(-, b)$ is a free right $\mathcal{D}$ -set and (G) free bases close into a wide subcategory.
- (T3) *The (G)-obstruction is an H^2 class* [2]. Assume (L) and hypothesis (H) below. Then the free $\mathcal{D}$ -orbits form a category $\text{Sk}_{\mathcal{C}}$, the vertex groups $G_a = \text{Aut}_{\mathcal{D}}(a)$ form a presheaf $\mathcal{D} : \text{Sk}_{\mathcal{C}}^{\text{op}} \rightarrow \mathbf{Ab}$, every transversal T has a defect 2-cocycle ω_T , regauging changes it by a coboundary, and (G) $\iff [\omega] := [\omega_T] = 0 \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$. The complex is Baues–Wirsching [6]; the equivalence with extension theory is classical.

Hypothesis (H) [2]. (i) $\mathcal{D}$ is a disjoint union of abelian vertex groups $G_a = \text{Aut}_{\mathcal{D}}(a)$, with $\mathcal{D}(a, b) = \emptyset$ for $a \neq b$; (ii) for every generator $c : a \rightarrow b$ and $\psi \in G_b$, $\psi \circ c$ lies in the $\mathcal{D}$ -orbit of c (trivial left action), i.e. $\psi \circ c = c \circ \varphi_c(\psi)$ for a unique $\varphi_c(\psi) \in G_a$.

Recall the mechanism of the acyclic examples [2, 3, 4]: the nontrivial vertex group sat at a *source*, so every restriction map φ_c vanished, $C^3 = 0$, and $H^2(\text{Sk}_{\mathcal{D}}; \mathcal{D})$ was a small finite group cut out by the branch geometry alone. *This source-concentration is exactly what a groupoid destroys*: in a groupoid every object receives arrows, so no vertex group sits at a pure source, and the restriction maps become isomorphisms.

Lemma 1 (Restrictions are isomorphisms over a groupoid). *If $\text{Sk}_{\mathcal{D}}$ is a groupoid then every restriction map $\varphi_c : G_b \rightarrow G_a$ ($c : a \rightarrow b$ in $\text{Sk}_{\mathcal{D}}$) is an isomorphism; hence $\mathcal{D}$ is a local system (a functor inverting every arrow), i.e. a module over the fundamental group(oid).*

Proof. φ is functorial and contravariant, $\varphi_{c_2 * c_1} = \varphi_{c_1} \circ \varphi_{c_2}$, with $\varphi_{\text{id}} = \text{id}$ [2, Lem. skel]. If c has a two-sided inverse c^{-1} in $\text{Sk}_{\mathcal{D}}$ then $\varphi_c \circ \varphi_{c^{-1}} = \varphi_{c^{-1} * c} = \varphi_{\text{id}} = \text{id}$ and symmetrically, so φ_c is invertible. $\square$

2 The one-object groupoid base is a group extension

The cleanest connected groupoid is a group Γ viewed as a one-object groupoid $B\Gamma$. We show the entire obstruction problem, over such a base, is the classical theory of group extensions. Throughout this section $\mathcal{K}$ is a *group* (a one-object groupoid) and $\mathcal{D} \leq \mathcal{K}$ a subgroup; we write the single object $*$ and identify arrows with group elements.

Proposition 2 (Orbits, hypothesis, and skeleton). *Let $\mathcal{K}$ be a group and $\mathcal{D} \leq \mathcal{K}$ an abelian subgroup. The right $\mathcal{D}$ -action on $\mathcal{K}$ by $f \cdot d = fd$ has orbits the left cosets $f\mathcal{D}$, and each orbit is free. Hence (L) holds. Hypothesis (H)(i) holds iff $\mathcal{D}$ is abelian; (H)(ii) holds iff $\mathcal{D}$ is normal in $\mathcal{K}$, and then the orbit category is the quotient group*

$$\text{Sk}_{\mathcal{D}} = \mathcal{K}/\mathcal{D} =: \Gamma,$$

a one-object connected groupoid, with presheaf $\mathcal{D}() = \mathcal{D}$ carrying the Γ -action by conjugation $\varphi_{[c]}(\psi) = c^{-1}\psi c$ (well defined on Γ because $\mathcal{D}$ is abelian and normal).*

Proof. Freeness of the right regular action on cosets is immediate ($fd = fd' \Rightarrow d = d'$). (H)(i) is the definition of abelian $\mathcal{D}$. For (H)(ii): the vertex group is $G_* = \mathcal{D}$ and a generator of the coset $c\mathcal{D}$ is c ; $\psi \circ c = \psi c$ lies in $c\mathcal{D}$ iff $c^{-1}\psi c \in \mathcal{D}$, i.e. iff c normalises $\mathcal{D}$. Requiring this for all coset representatives c is normality of $\mathcal{D}$. When $\mathcal{D} \trianglelefteq \mathcal{K}$, orbit composition $c_2\mathcal{D} * c_1\mathcal{D} = \text{orb}(c_2c_1) = c_2c_1\mathcal{D}$ is well defined (it is coset multiplication in $\mathcal{K}/\mathcal{D}$), giving $\text{Sk}_{\mathcal{D}} = \Gamma$; and conjugation descends to a Γ -action on $\mathcal{D}$ because $\mathcal{D}$ is central-by-nothing—abelian and normal makes $\mathcal{D}$ a Γ -module. Contravariance $\varphi_{[c]}(\psi) = c^{-1}\psi c$ matches Lemma 1. $\square$

Proposition 3 (The defect is the Schreier factor set). *With $\mathcal{K}, \mathcal{D}, \Gamma$ as in Proposition 2, choose a transversal $T = \{c_\gamma : \gamma \in \Gamma\}$ of coset representatives with $c_1 = \text{id}$ (the unit normalization). The defect 2-cochain of (T3) is*

$$c_{\gamma_2} c_{\gamma_1} = c_{\gamma_2 \gamma_1} \cdot \omega_T(\gamma_2, \gamma_1), \quad \omega_T(\gamma_2, \gamma_1) \in \mathcal{D}, \quad (1)$$

i.e. ω_T is exactly the factor set (Schreier 2-cocycle) of the group extension

$$1 \longrightarrow \mathcal{D} \longrightarrow \mathcal{K} \xrightarrow{\pi} \Gamma \longrightarrow 1, \quad \pi = \text{quotient by } \mathcal{D}. \quad (2)$$

Consequently $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) = H^2(\Gamma; \mathcal{D})$ is the extension class of (2), and:

$$\mathcal{K} = \mathcal{C} \bowtie \mathcal{D} \text{ (the merge exists)} \iff (2) \text{ splits} \iff \mathcal{D} \text{ has a complement in } \mathcal{K} \iff [\omega] = 0.$$

Moreover the closing transversals (the strict factorizations) are exactly the group-theoretic complements of $\mathcal{D}$, so

$$\#\{\text{strict factorizations}\} = \#\{\text{complements of } \mathcal{D} \text{ in } \mathcal{K}\}.$$

Proof. Equation (1) is Definition (defect) of (T3) specialised to $\mathcal{K}$ a group: the chosen generator $[c_{\gamma_2}c_{\gamma_1}]_T$ of the orbit $\text{orb}(c_{\gamma_2}c_{\gamma_1}) = (\gamma_2\gamma_1)\mathcal{D}$ is $c_{\gamma_2\gamma_1}$, and the residual $\mathcal{D}$ -element is ω_T . This is verbatim the defining relation of a factor set of (2) [7, 8]. The cocycle identity (1) pushed through associativity is the standard 2-cocycle condition in $H^2(\Gamma; \mathcal{D})$ with the conjugation action of Proposition 2; regauging $c_\gamma \mapsto c_\gamma h_\gamma$ ($h_\gamma \in \mathcal{D}$) is the standard coboundary. By (T3), $[\omega] = 0 \iff$ (G), and by (T2) with (L), (G) $\iff \mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$. A transversal closes iff $\omega_T \equiv \text{id}$ iff the c_γ satisfy $c_{\gamma_2}c_{\gamma_1} = c_{\gamma_2\gamma_1}$, i.e. iff $\{c_\gamma\}$ is a subgroup mapping isomorphically onto Γ — a complement. Splitting of (2) $\iff [\omega] = 0$ is Schreier’s theorem [7, 8]. $\square$

Proposition 3 is the conceptual pivot: *over a one-object groupoid base and under (H), the Zappa–Szép merge obstruction and the group-extension obstruction are the same class.* The Zappa–Szép product $\mathcal{C} \bowtie \mathcal{D}$ with $\mathcal{D}$ normal is the internal semidirect product $\mathcal{C} \ltimes \mathcal{D}$; its existence is a splitting; the holonomy is Schreier’s H^2 .

3 Refutation: a connected groupoid base with nonzero obstruction

Theorem 4 (Invertibility does not trivialise the obstruction). *There is a directed container presenting the connected groupoid $B(\mathbb{Z}/2)$ and a right factor $\mathcal{D} \cong \mathbb{Z}/2$ of internal relabelings for which $[\omega] \neq 0$; hence no Zappa–Szép product / merge exists. Explicitly, take $\mathcal{K} = \mathbb{Z}/4$ and $\mathcal{D} = \{0, 2\} \cong \mathbb{Z}/2$. Then $\text{Sk}_{\mathcal{C}} = \mathbb{Z}/4 / \mathbb{Z}/2 = \mathbb{Z}/2 = B(\mathbb{Z}/2)$ is a connected groupoid, (L) and (H) hold, and*

$$[\omega] = \text{the generator of } H^2(\mathbb{Z}/2; \mathbb{Z}/2) \cong \mathbb{Z}/2,$$

the class of the non-split extension $1 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z}/4 \rightarrow \mathbb{Z}/2 \rightarrow 1$.

Proof. $\mathbb{Z}/4$ is abelian so $\mathcal{D} = \{0, 2\}$ is abelian and normal; by Propositions 2–3, (L) and (H) hold, $\text{Sk}_{\mathcal{C}} = \mathbb{Z}/2$, the action of $\mathbb{Z}/2$ on $\mathcal{D}$ is trivial (abelian $\mathcal{K}$), and $[\omega]$ is the extension class of $\mathbb{Z}/4$. The only order-2 subgroup of $\mathbb{Z}/4$ is $\mathcal{D}$ itself, so $\mathcal{D}$ has *no* complement: the extension is non-split and $[\omega] \neq 0$. Since $H^2(\mathbb{Z}/2; \mathbb{Z}/2) \cong \mathbb{Z}/2$ (trivial action), $[\omega]$ is its generator. All of this is machine-checked (§6, example (B)): $\#\text{complements} = 0$, the $\mathbb{F}_2$ bar complex has $\dim H^2 = 1$, and $\omega_T \notin B^2$. $\square$

Remark 5 (Where the conjecture broke). The seed conjecture read “connected groupoid $\Rightarrow \text{Sk}_{\mathcal{C}}$ contractible $\Rightarrow H^2 = 0$.” A connected groupoid $\text{Sk}_{\mathcal{C}}$ is equivalent, as a category, to its vertex group Γ ; its classifying space is $B\Gamma = K(\Gamma, 1)$, which is contractible iff $\Gamma = 1$. For $\Gamma = \mathbb{Z}/2$, $B\Gamma = \mathbb{RP}^\infty$ has $H^2(\mathbb{RP}^\infty; \mathbb{Z}/2) = \mathbb{Z}/2 \neq 0$. The broken assumption was the tacit identification of “every object isomorphic to every other” (which *does* hold in a connected groupoid, making the *skeleton* a single object) with “the classifying space is a point” (which needs the vertex group trivial too). Invertibility collapses the objects but not the loops; the loops are precisely the $H^{\geq 1}$.

Theorem 6 (Same base, split total: obstruction is genuinely about $\mathcal{K}$). *The value of $[\omega]$ is not a function of the base groupoid alone. With the same base $\text{Sk}_{\mathcal{C}} = B(\mathbb{Z}/2)$ and the same coefficient $\mathcal{D} \cong \mathbb{Z}/2$, the total category $\mathcal{K} = \mathbb{Z}/2 \times \mathbb{Z}/2$ with $\mathcal{D} = \langle (1, 0) \rangle$ has $[\omega] = 0$: $\mathcal{D}$ has two complements $\langle (0, 1) \rangle, \langle (1, 1) \rangle$, the split extension merges.*

Proof. Machine-checked (§6, example (C)): $\#\text{complements} = 2$, $\omega_T \in B^2$, $[\omega] = 0$. Analytically, $\mathbb{Z}/2 \times \mathbb{Z}/2$ is the split extension of $\mathbb{Z}/2$ by $\mathbb{Z}/2$, whose class is 0. $\square$

A larger, non-cyclic groupoid base obstructs in the same way.

Theorem 7 (A rank-two groupoid base). *Let $\mathcal{K} = Q_8$ (quaternion group) and $\mathcal{D} = Z(Q_8) = \{\pm 1\} \cong \mathbb{Z}/2$. Then $\text{Sk}_{\mathcal{C}} = Q_8/\{\pm 1\} = (\mathbb{Z}/2)^2$ is a connected groupoid, (H) holds, $H^2((\mathbb{Z}/2)^2; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^3$, and $[\omega] \neq 0$: every subgroup of Q_8 contains -1 , so $\mathcal{D}$ has no complement and the extension is non-split. No merge exists.*

Proof. Machine-checked (§6, example (D)): the $\mathbb{F}_2$ bar complex returns $\dim H^2 = 3$ (so $|H^2| = 8$, the known $H^*((\mathbb{Z}/2)^2; \mathbb{Z}/2) = \mathbb{Z}/2[x, y]$ with $\dim H^2 = 3$), $\#\text{complements} = 0$, and $\omega_T \notin B^2$. Q_8 is the non-split central extension realising the class $x^2 + xy + y^2$. $\square$

4 The correct positive theorem: cohomological dimension, not invertibility

The conjecture’s *conclusion* — “these bases always compose” — is true for the cases that matter, but for a reason invisible to the contractibility story.

Theorem 8 ($\text{cd} \leq 1$ forces the merge, for every $\mathcal{D}$). *Suppose the skeleton $\text{Sk}_{\mathcal{C}}$ has cohomological dimension ≤ 1 , i.e. $H^2(\text{Sk}_{\mathcal{C}}; A) = 0$ for every coefficient system $A : \text{Sk}_{\mathcal{C}}^{\text{op}} \rightarrow \mathbf{Ab}$. Then for every right factor $\mathcal{D}$ satisfying (L) and (H), $[\omega] = 0$ and the Zappa–Szép product $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ exists. In particular this holds whenever $\text{Sk}_{\mathcal{C}}$ is a free groupoid — every connected component has a free vertex group — e.g. the free groupoid on any graph.*

Proof. By (T3) the class $[\omega]$ lives in $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$, which is 0 by hypothesis; then (G) holds and (T2) gives the product. For the free-groupoid case: a connected groupoid is equivalent to its vertex group Γ , and Baues–Wirsching cohomology is invariant under equivalence of categories [6], so $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \prod_{\text{comp}} H^2(\Gamma_i; M_i)$ (group cohomology, M_i the stalk, a Γ_i -module by Lemma 1). A free group has cohomological dimension ≤ 1 (Stallings–Swan [9, 10]: $\text{cd } \Gamma \leq 1 \iff \Gamma$ free), whence $H^{\geq 2}(\Gamma_i; M) = 0$ for every module M . Thus $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) = 0$. $\square$

Corollary 9 (Trichotomy for a connected groupoid base). *Let $\text{Sk}_{\mathcal{C}}$ be a connected groupoid with vertex group Γ and let $\mathcal{D}$ satisfy (H), giving a Γ -module M and an extension $1 \rightarrow M \rightarrow \mathcal{K} \rightarrow \Gamma \rightarrow 1$. Then $[\omega] \in H^2(\Gamma; M)$ is its class, and:*

- (a) Γ **free** ($\text{cd} \leq 1$): $H^2(\Gamma; M) = 0$ for all M ; the merge always exists.
- (b) Γ **trivial** ($\Gamma = 1$): $\text{Sk}_{\mathcal{C}}$ is genuinely contractible (the seed conjecture’s true subcase); the merge exists. This is (a) with Γ free of rank 0, and is realised by codiscrete groupoids (§6, example (A)).
- (c) Γ **has torsion**: $H^2(\Gamma; M)$ can be nonzero (Theorems 4, 7); the merge exists iff the specific extension splits.

The dividing line is $\text{cd}(\Gamma)$, not invertibility. Freeness — the absence of holonomy loops beyond dimension one — is what trivialises the obstruction; a non-free (torsion) symmetry, even though fully invertible, can carry a nonzero degree-two class.

Remark 10 (Unification with the acyclic examples). Both regimes are the *same* invariant $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$; the base’s homotopy type chooses the mechanism. Over an *acyclic* base (a poset/DAG, the free category on an acyclic quiver) the nontrivial vertex group sits at a source, the restriction maps vanish, and H^2 is cut out by branch-and-rejoin geometry — the $\mathbb{Z}/2$ rigid twist, the $\mathbb{Z}/n$ lot-cursor [2, 3, 4]. Over a *groupoid* base every restriction map is an isomorphism (Lemma 1) and H^2 is group cohomology of Γ . The extremes meet in one slogan: the merge is automatic exactly when $\text{Sk}_{\mathcal{C}}$ is one-dimensional ($\text{cd} \leq 1$) — a forest of handoffs, or a free groupoid of reversible ties — and it can obstruct the moment the handoff skeleton acquires a genuine degree-two cocycle, whether from a rejoining acyclic branch or from a torsion loop.

5 Application: which social networks merge for free

The result stages Neil’s social-network exploration. Model a platform as a directed container whose skeleton is the free (categorical or groupoidal) structure on its tie-graph; merging two platforms sharing users is the Zappa–Szép composition $\mathcal{C} \bowtie \mathcal{D}$ along the shared sub-structure $\mathcal{D}$ of relabelings/identifications. The grading here is COMPUTED: “a real network *is* this groupoid” is a faithful abstraction, not a fidelity claim (the standing SEED-Q4 caveat).

Corollary 11 (Mutual ties merge; the reason is dimension one). *A network of mutual (symmetric) ties presents the free groupoid on its friendship graph. Its vertex groups are the fundamental groups of the graph’s connected components — free groups (a graph is a $K(\text{free}, 1)$). By Theorem 8, $H^2 = 0$ for every coefficient system: mutual-tie platforms always merge consistently, for any right factor of shared identifications. By contrast a network of directed (follows) ties presents a free category on a directed quiver; when it is acyclic the earlier analysis applies and a rejoining of two follow-paths can obstruct ($[\omega] = \varepsilon \in \mathbb{Z}/n$, [3, 4]). And a shared symmetry with torsion — a finite cyclic identification cycle of order n that is not free — can obstruct even in the mutual case (Theorem 4 is $n = 2$).*

The corrected headline for the grant’s Impact line is therefore sharper than the conjecture: *reversibility alone does not guarantee that two platforms merge; **freeness** does*. Mutual-tie graphs merge because their groupoids are one-dimensional, not because they are trivial; introduce a torsion identification (a quotient that forces a nontrivial finite loop) and the merge can fail with a computable degree-two obstruction.

6 Computational verification

All finite claims are machine-checked in `projects/scratch/groupoid-zs/`: `groupoid_zs.py` (orbits, (L), (H), $\text{Sk}_{\mathcal{C}}$, the defect, `#complements`, and the $\mathbb{F}_2$ bar complex with the *correct* restriction maps φ), `dcont_check.py` (D1–D5 for the bases). The φ computation matters: unlike the acyclic scripts, the groupoid restriction maps are isomorphisms, not zero (Lemma 1), so the earlier hard-coded $\varphi = 0$ is replaced by conjugation.

	base $\text{Sk}_{\mathcal{C}}$	$\mathcal{D}$	(L),(H)	$\dim H^2$	#compl.	verdict
(A) codiscrete $\bullet \rightrightarrows \bullet$	1 (contractible)	{id}	yes	0	1	merges
(B) $\mathbb{Z}/4$	$B(\mathbb{Z}/2)$	$\mathbb{Z}/2$	yes	1	0	obstructed
(C) $\mathbb{Z}/2 \times \mathbb{Z}/2$	$B(\mathbb{Z}/2)$	$\mathbb{Z}/2$	yes	1	2	merges
(D) Q_8	$B((\mathbb{Z}/2)^2)$	$\mathbb{Z}/2$	yes	3	0	obstructed
(E) $\mathbb{Z}/6$	$B(\mathbb{Z}/2)$	$\mathbb{Z}/3$	yes	—	1	merges

Examples (B),(D) are connected-groupoid bases with $[\omega] \neq 0$ (Theorems 4,7); (C) is the same base as (B) with a split total (Theorem 6); (E) illustrates Schur–Zassenhaus (coprime $|\mathcal{D}|, |\Gamma|$ force a split, $[\omega] = 0$). In every case $\#\text{complements} > 0 \iff [\omega] = 0$, confirming Proposition 3; $\dim H^2$ reproduces the known group cohomologies $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$, $H^2((\mathbb{Z}/2)^2; \mathbb{Z}/2) = (\mathbb{Z}/2)^3$; and $\delta^2\delta^1 = 0$ is verified on the full (now non-vacuous) C^3 .

7 Status, novelty, and honest scope

What is proved. (i) The reduction: over a one-object groupoid base under (H), the merge obstruction is exactly the Schreier extension class, and the merge exists iff the extension splits (Propositions 2–3). (ii) The refutation of the strong conjecture: connected-groupoid bases $B(\mathbb{Z}/2)$ and $B((\mathbb{Z}/2)^2)$ with $[\omega] \neq 0$ (Theorems 4,7). (iii) The correct positive theorem: $\text{cd}(\text{Sk}_{\mathcal{C}}) \leq 1$ — in particular a free groupoid — forces the merge for every $\mathcal{D}$ (Theorem 8), giving the social-network corollary with the right reason (Corollary 11). Grade: the worked *instances* are COMPUTED (machine-checked); the reduction and the $\text{cd} \leq 1$ theorem are PROVED, resting on the cited (T2), (T3), Schreier theory, and Stallings–Swan.

What is *not* claimed (novelty firewall). Group extensions and their H^2 classification are classical (Schreier; [7, 8]); the identification $H^2_{\text{BW}}(B\Gamma; M) = H^2(\Gamma; M)$ is standard. Groupoid Zappa–Szép products themselves exist in the operator-algebra / self-similar literature (Mundey–Sims and neighbours), on the *completeness* axis, and are cited as adjacent, *not* a scoop. The contribution here is narrow and precise: placing the groupoid case inside the Zappa–Szép merge program, *refuting* the seed contractibility conjecture, and isolating $\text{cd} \leq 1$ (freeness) as the true trivialising condition — one honest correction, machine-witnessed, that redirects the application story from “reversible $\Rightarrow$ mergeable” to “free $\Rightarrow$ mergeable.”

Open. The nonabelian regime ((H)(ii) fails: $\mathcal{D}$ non-normal, i.e. a genuine two-sided Zappa–Szép matched pair rather than a semidirect product) pushes the obstruction to a nonabelian H^2 of the matched pair (Kac/Pirashvili, cited in [2]); we do not enter it. And the fidelity question — whether a given real network *is* the free groupoid on its tie-graph — stays OPEN (SEED-Q4).

References

- [1] MacBeth, *The pairwise Zappa–Szép criterion*, Kodamai note, 10 June 2026.
- [2] MacBeth, *The global-closure obstruction (G) is a cohomology class: the Zappa–Szép holonomy as $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$* , Kodamai note, 11 June 2026.
- [3] MacBeth, *The re-entrancy obstruction is the token-mutation bit*, Kodamai note, 20 July 2026.
- [4] MacBeth, *Supply-chain composition is a Zappa–Szép product*, Kodamai note, 23 July 2026.

- [5] R. Rosebrugh, R. J. Wood, *Distributive laws and factorization*, J. Pure Appl. Algebra 175 (2002), 327–353.
- [6] H.-J. Baues, G. Wirsching, *Cohomology of small categories*, J. Pure Appl. Algebra 38 (1985), 187–211.
- [7] S. Mac Lane, *Homology*, Springer, 1963 (Ch. IV: group extensions and H^2).
- [8] K. S. Brown, *Cohomology of Groups*, GTM 87, Springer, 1982.
- [9] J. Stallings, *On torsion-free groups with infinitely many ends*, Ann. of Math. 88 (1968), 312–334.
- [10] R. G. Swan, *Groups of cohomological dimension one*, J. Algebra 12 (1969), 585–610.